Current-Aware Rendezvous Planning for
Multiple Patrolling USVs with Heading Constraints

Motivation & Problem

- Efficient rendezvous planning is required to inspect multiple patrolling USVs, particularly when different priorities are assigned to individual USVs.
- Ocean currents and heading constraints make travel time dependent on both position and direction, distinguishing this problem from conventional rendezvous planning.
- This study proposes an integrated framework that combines constrained path planning with learning-based rendezvous optimization.

Key Contributions

- A graph-based path-planning method using multi-stage maneuver approximation to account for ocean currents and heading constraints
- A graph neural network architecture that encodes spatiotemporal relationships among rendezvous candidates for priority-aware planning

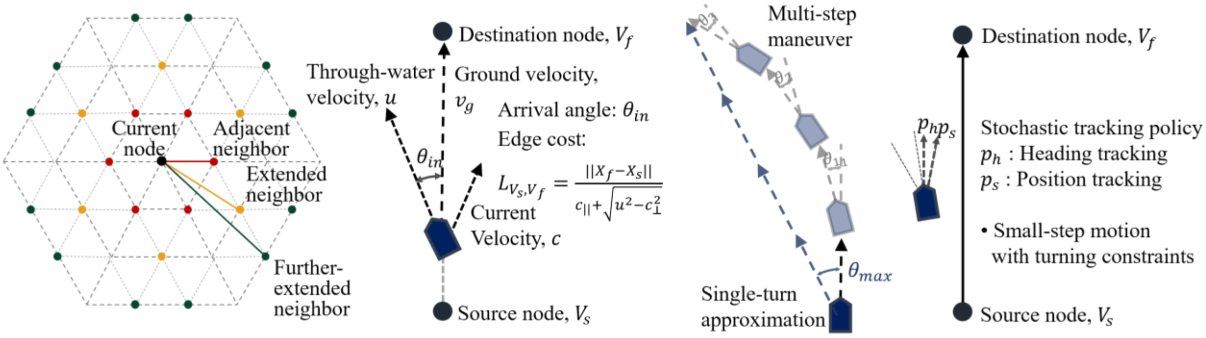
PROPOSED APPROACH

Current-Aware Navigation

• Grid graph-based navigation for large maritime areas
• Current-aware edge costs with heading-constrained connectivity
• Multi-stage maneuver approximation for large heading changes
• Hybrid path tracking for feasible inter-node turning angles, θ_{max}

Rendezvous Planning

• Represent discretized rendezvous points as graph nodes.
• Earliest feasible transitions between rendezvous points of different USVs
• Gated GCN encoding with travel time and due date as edge features
• PPO-based policy learning with a decision tree surrogate for accelerated travel-time estimation



(a) Grid graph (b) Edge cost computation (c) Multi step maneuver approximation (d) Stochastic path validation

Figure 1. Current-Aware Path Planning

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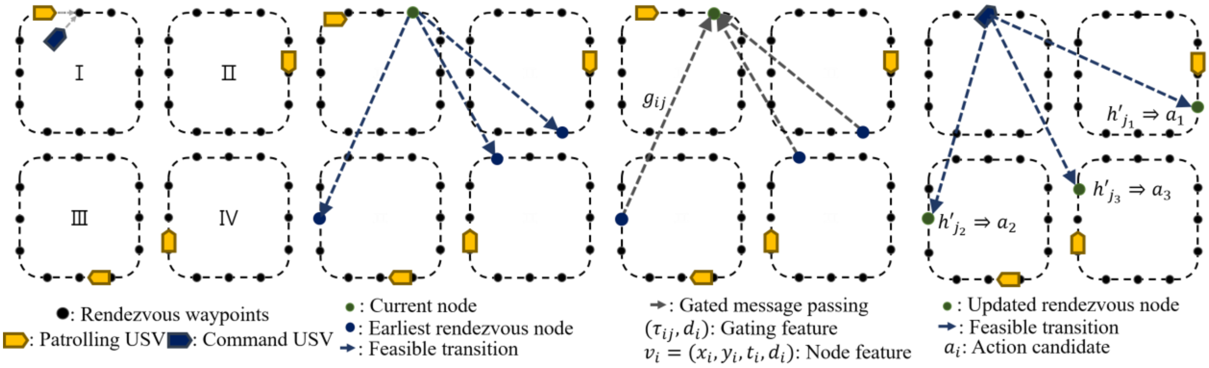


(a) Rendezvous Scenario (b) Feasible Transition Graph (c) Gated Message Aggregation (d) Rendezvous Action Representation

Figure 2 Rendezvous Planning

Key Result

Table 1. Average scaled objective value of each algorithm

Algorithm	12-2T	12-3T	12-4T	16-2T	16-3T	16-4T	24-2T	24-3T	24-4T
Graph-RL	3.72 ± 0.7	**4.16 ± 0.8**	4.44 ± 1.1	**4.39 ± 0.6**	**5.56 ± 1.2**	6.02 ± 1.4	**7.1 ± 2.2**	**7.62 ± 1.5**	**8.92 ± 1.9**
MLP-RL	3.80 ± 0.7	4.42 ± 0.9	4.62 ± 1.0	4.66 ± 0.6	5.79 ± 1.0	6.24 ± 1.3	7.81 ± 2.3	8.31 ± 1.3	9.43 ± 1.5
Greedy	4.10 ± 1.0	4.79 ± 1.3	5.32 ± 1.5	5.15 ± 1.2	6.53 ± 1.6	7.02 ± 1.8	9.53 ± 3.7	10.2 ± 2.6	11.6 ± 2.6
PCNN	3.86 ± 0.6	4.78 ± 1.5	6.40 ± 2.3	4.67 ± 0.9	6.52 ± 1.8	9.05 ± 3.6	8.67 ± 4.3	9.92 ± 3.2	15.5 ± 6.8
Rolling Greedy	3.83 ± 0.7	4.27 ± 1.0	4.85 ± 1.2	4.79 ± 0.9	5.93 ± 1.5	6.43 ± 1.5	8.15 ± 2.0	8.40 ± 1.7	9.43 ± 1.7

Table 2. Relative Improvement over Baseline Methods (%)

Baseline Comparison	12-2T	12-3T	12-4T	16-2T	16-3T	16-4T	24-2T	24-3T	24-4T	Total
Graph-MLP	1.21%	4.53%	2.84%	5.15%	3.38%	2.55%	8.39%	7.14%	4.87%	4.45%
p-value	0.340	0.0079	0.124	0.0008	0.0818	0.149	0.0001	0.0018	0.0207	< 0.001

* N-MT: problem setting with N patrolling USVs and M Tier priority distribution

* PCNN: Priority-Constrained Nearest-Neighbor rule, * Rolling greedy: 2 steps rolling greedy

* Dark/light gray: performance advantage at the 5%/10% significance level

• USV priorities were represented using tiered due dates, and the objective minimized total mission time and cumulative tardiness.

• The methods were evaluated on 60 fixed instances with randomized priorities, initial positions, and patrol directions across different fleet sizes and patrol configurations.

• Graph-RL is the proposed GNN-based method, while MLP-RL is a strong learning-based heuristic using only local USV information.

• Graph-RL generalized well beyond its 12-3T training distribution and showed increasingly clear advantages over MLP-RL and the greedy baseline as the fleet size increased, particularly in the 24-USV cases.

• The results suggest that explicitly modeling the relationships among spatiotemporal rendezvous coordinates becomes increasingly effective as the number of feasible alternatives grows.

Reward formulation ablation

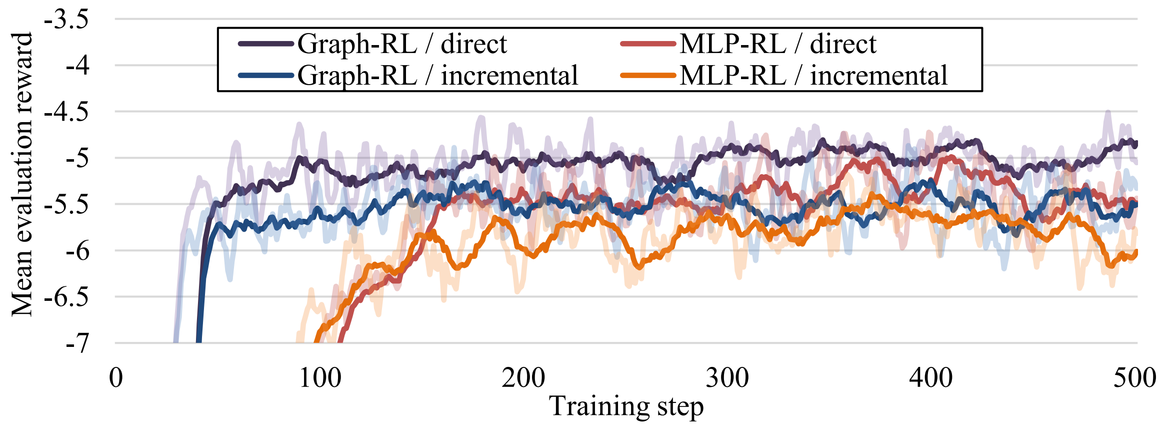


Figure 3 Reward Formulation Ablation

• The tardiness reward can be formulated in two ways: direct and incremental. In the direct formulation, only the tardiness associated with the selected USV is penalized. In the incremental formulation, the reward accounts for the total increase in tardiness across all unvisited USVs during each decision step.

• For both Graph-RL and MLP-RL, the direct tardiness formulation achieved better performance. Graph-RL also exhibited faster convergence and smaller training fluctuations.

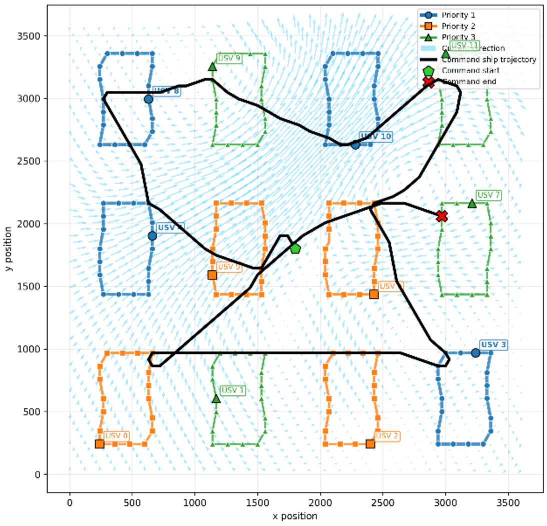
Discussion

• Figure 4 presents representative route solutions generated by Graph-RL and MLP-RL.

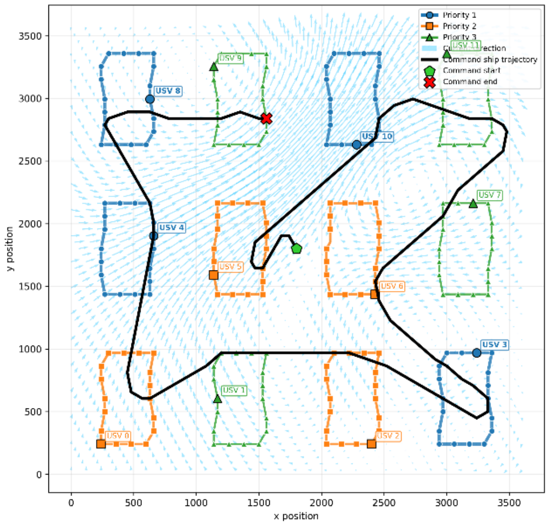
* Priority 1 denotes the highest priority, whereas Priority 3 denotes the lowest.

• Both methods produce high-quality routes that prioritize high-priority USVs while opportunistically rendezvousing with multiple lower-priority USVs along the way.

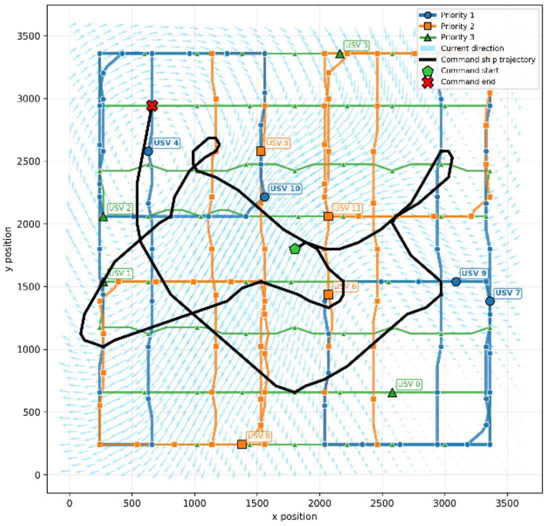
• By modeling the spatial relationships among all USVs rather than relying solely on local information, Graph-RL generates routes that reflect both the immediate outcome of the current action and its implications for subsequent rendezvous decisions.



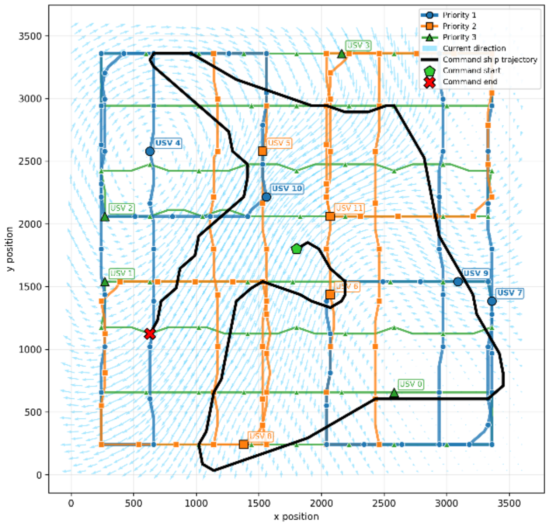
Graph-RL (12-3T-a)



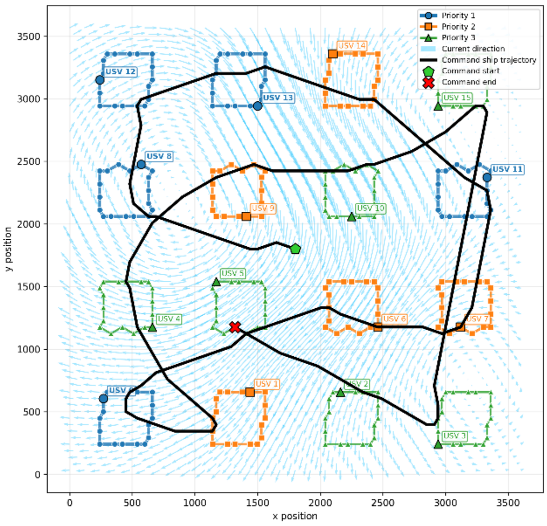
MLP-RL (12-3T-a)



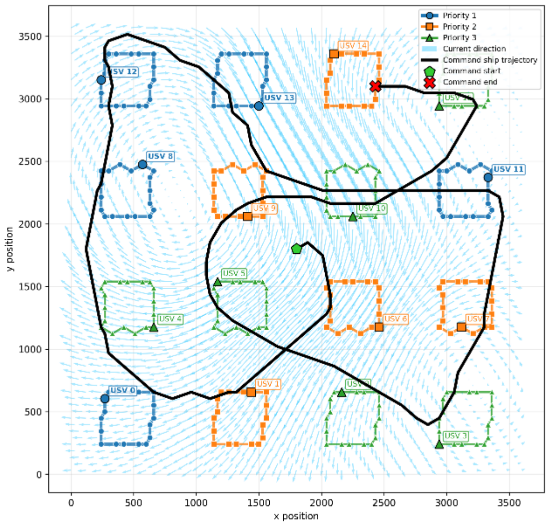
Graph-RL (12-3T-b)



MLP-RL (12-3T-b)



Graph RL (16-3T-a)



MLP-RL (16-3T-a)

Figure 4. Rendezvous Route Comparison: Graph-RL vs. MLP-RL